UE26MA141A · ENGINEERING MATHEMATICS – I · SHORT NOTES

Unit 2 — Higher Order Linear Differential Equations

Last-minute revision: CF rules, every PI type with its failure case, variation of parameters, Cauchy and Legendre equations, LCR circuits and SHM, with worked answers. Built from the PESU Unit 2 slides (sessions 1–11).

1. Linear DEs: definitions & operator D
2. Complementary function (CF)
3. Inverse operator & particular integral (PI)
4. PI Type 1: $X = e^{ax}$
5. PI Type 2: $X = \sin(ax + b), \cos(ax + b)$
6. PI Type 3: $X = x^m$
7. PI Type 4: $X = e^{ax}V(x)$
8. PI when X is any other function
9. Variation of parameters
10. Cauchy's & Legendre's equations
11. Applications: LCR circuit & SHM
12. One-page formula sheet

1 Linear Differential Equations — Basics

A DE is **linear** if the dependent variable y and its derivatives appear only to the first degree and are never multiplied together.

N-TH ORDER LINEAR DE WITH CONSTANT COEFFICIENTS

$$\frac{d^n y}{dx^n} + a_1\frac{d^{n-1}y}{dx^{n-1}} + \ldots + a_{n-1}\frac{dy}{dx} + a_n y = X(x)$$

With $\boldsymbol{D = d/dx}$ (so $D^2 = d^2/dx^2$, ...) this becomes $\boldsymbol{f(D)\,y = X}$, where $f(D) = D^n + a_1D^{n-1} + \ldots + a_n$ is the linear differential operator.

Type	Condition	Solution
Homogeneous	$X = 0$	$y = CF$
Non-homogeneous	$X \neq 0$	$\boldsymbol{y = CF + PI}$ (complete or general solution)

- **CF** (complementary function) is the general solution of $f(D)y = 0$ and contains n arbitrary constants.
- **PI** (particular integral) is any one solution of $f(D)y = X$ and contains **no** arbitrary constants.
- **Superposition:** if y_1 and y_2 solve the homogeneous equation, so does $c_1y_1 + c_2y_2$.
- y_1 and y_2 are **linearly independent** ⇔ the Wronskian $W = y_1y_2' - y_2y_1' \neq 0$.

2 Complementary Function (CF)

Replace D by m to get the **auxiliary equation (AE)** $f(m) = 0$, find its roots, and read off the CF:

Roots of AE	CF
Real & distinct: $m_1, m_2, \ldots$	$c_1e^{m_1x} + c_2e^{m_2x} + \ldots$
Real & repeated: m, m (twice)	$(c_1 + c_2x)e^{mx}$
Repeated k times	$(c_1 + c_2x + \ldots + c_kx^{k-1})e^{mx}$
Complex: $\alpha \pm i\beta$	$e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$
Complex repeated: $\alpha \pm i\beta$ (twice)	$e^{\alpha x}[(c_1 + c_2x)\cos \beta x + (c_3 + c_4x)\sin \beta x]$
Pure imaginary: $\pm i\beta$	$c_1 \cos \beta x + c_2 \sin \beta x$

CF examples

$(D^3 - 6D^2 + 11D - 6)y = 0$	$m = 1, 2, 3$ (try $m = 1$ first, then synthetic division)	$c_1e^x + c_2e^{2x} + c_3e^{3x}$
$(D^2 + 2D + 5)y = 0$	$m = -1 \pm 2i$	$e^{-x}(c_1 \cos 2x + c_2 \sin 2x)$
$(D^2 - 2D + 1)y = 0$	$m = 1, 1$	$(c_1 + c_2x)e^x$
$(D^4 + 8D^2 + 16)y = 0$	$(m^2 + 4)^2 = 0 \Rightarrow \pm 2i$ twice	$(c_1 + c_2x)\cos 2x + (c_3 + c_4x)\sin 2x$
$(2D^2 + 2D + 3)y = 0$	$m = (-1 \pm i\sqrt{5})/2$	$e^{-x/2}(c_1 \cos \frac{\sqrt{5}}{2}x + c_2 \sin \frac{\sqrt{5}}{2}x)$
$(D^4 - 1)y = 0$	$m = \pm 1, \pm i$	$c_1e^x + c_2e^{-x} + c_3 \cos x + c_4 \sin x$

3 Inverse Operator & Particular Integral

$$\mathbf{PI} = \frac{1}{f(D)}X \qquad \frac{1}{D}X = \int X\,dx \qquad \frac{1}{D-a}X = e^{ax}\int X\,e^{-ax}dx \qquad \frac{1}{D+a}X = e^{-ax}\int X\,e^{ax}dx$$

$1/f(D)$ is the inverse of $f(D)$: $f(D)[1/f(D)]X = X$. The shortcut rules below follow from $D\,e^{ax} = a\,e^{ax}$ and $D^2 \sin ax = -a^2 \sin ax$.

4 PI Type 1 — $X = e^{ax}$ (also constants, sinh, cosh, k^x)

$$\frac{1}{f(D)}e^{ax} = \frac{e^{ax}}{f(a)}, \quad f(a) \neq 0 \quad \text{(replace } D \text{ by } a\text{)}$$

Case of failure ($f(a) = 0$, i.e. a is a root of the AE):

$$\text{PI} = x\frac{e^{ax}}{f'(a)} \quad (f'(a) \neq 0); \qquad \text{if } f'(a) = 0 \text{ too: PI} = x^2\frac{e^{ax}}{f''(a)}; \qquad \text{and so on.}$$

- **Constant** $k = k\,e^{0x} \Rightarrow$ put $D = 0$.
- $\sinh ax = (e^{ax} - e^{-ax})/2$ and $\cosh ax = (e^{ax} + e^{-ax})/2$: split and apply the rule to each part.
- $k^x = e^{x \ln k} \Rightarrow a = \ln k$. For a sum $X = X_1 + X_2$, $\text{PI} = \text{PI}_1 + \text{PI}_2$.

Equation	Working	PI
$(D^2 + 2D + 5)y = 1$	$X = e^{0x}, f(0) = 5$	**1/5**
$(D^2 + 4D + 4)y = e^{-2x}$	$f(-2) = 0, f'(-2) = 2D + 4 = 0, f'' = 2$	$\mathbf{x^2e^{-2x}/2}$
$(D^3 - 3D + 2)y = 2 \sinh x = e^x - e^{-x}$	$a = 1$: $f = f' = 0, f'' = 6D = 6 \Rightarrow x^2e^x/6$. $a = -1$: $f(-1) = 4$	$\mathbf{\frac{x^2e^x}{6} - \frac{e^{-x}}{4}}$
$(D^2 - 5D + 6)y = e^{3x} + 3$	$f(3) = 0, f'(3) = 2 \cdot 3 - 5 = 1$; constant: $f(0) = 6$	$\mathbf{x\,e^{3x} + ½}$

5 PI Type 2 — $X = \sin(ax+b)$ or $\cos(ax+b)$

$$\frac{1}{f(D^2)}\sin(ax+b) = \frac{\sin(ax+b)}{f(-a^2)},\ f(-a^2)\neq 0 \quad (\text{replace } \boldsymbol{D^2} \text{ by } \boldsymbol{-a^2};\ \text{the same rule holds for cos})$$

Case of failure ($f(-a^2)=0$): $\text{PI} = x\cdot\frac{1}{f'(D)}\sin(ax+b)$, then replace D^2 by $-a^2$ again. Most used special case:

$$\frac{1}{D^2+a^2}\sin ax = -\frac{x}{2a}\cos ax \qquad \frac{1}{D^2+a^2}\cos ax = \frac{x}{2a}\sin ax$$

- If odd powers of D remain after substitution (e.g. $1/(pD+q)$), multiply above and below by the **conjugate** $(pD-q)$ to get D^2 in the denominator, replace D^2 by $-a^2$, then let D act (differentiate) on the numerator.
- Rewrite powers and products first: $\sin^2x = (1-\cos 2x)/2$, $\cos^2x = (1+\cos 2x)/2$, $\sin A\cos B = \tfrac{1}{2}[\sin(A+B)+\sin(A-B)]$, $\sin^3x = (3\sin x - \sin 3x)/4$, $\cos^3x = (3\cos x + \cos 3x)/4$.

Equation	Working	PI
$(D^2+3)y = \cos\sqrt{3}x$	$f(-3)=0 \Rightarrow x\cdot(1/2D)\cos\sqrt{3}x = (x/2)\int\cos\sqrt{3}x\,dx$	$\frac{x\sin\sqrt{3}x}{2\sqrt{3}}$
$(D^3-2D^2+D)y = \sin^2x$	$X = \tfrac{1}{2}-\tfrac{1}{2}\cos 2x$. Part 1: $f(0)=0, f'(0)=1 \Rightarrow x/2$. Part 2: $D^2=-4$ gives $D^3=-4D$, denominator $8-3D$; $\times(8+3D) \Rightarrow (8+3D)\cos 2x/100$	$\frac{x}{2} - \frac{4\cos 2x - 3\sin 2x}{100}$
$(D^2+4)y = \sin 2x$	failure case with $a=2$	$-(x/4)\cos 2x$
$(D^2+D+1)y = \sin 2x$	$D^2=-4 \Rightarrow 1/(D-3) = (D+3)/(D^2-9) = (D+3)/(-13)$	$-(2\cos 2x + 3\sin 2x)/13$

6 PI Type 3 — $X = x^m$ (polynomial)

$\text{PI} = [f(D)]^{-1}x^m$ — take the lowest-degree term out, write the rest as $[1\pm\varphi(D)]^{-1}$, expand in ascending powers of D **up to D^m**, then operate term by term.

$(1+z)^{-1} = 1 - z + z^2 - z^3 + \ldots$ $\quad (1-z)^{-1} = 1 + z + z^2 + z^3 + \ldots$ $\quad (1+z)^{-2} = 1 - 2z + 3z^2 - 4z^3 + \ldots$

$(1-z)^{-2} = 1 + 2z + 3z^2 + 4z^3 + \ldots$ $\quad$ If $f(D)$ has no constant term, take out D: $\frac{1}{D(\ldots)}$ means integrate at the end.

Example 1 — $(D^2+3D+2)\,y = x^3+x^2$

CF: $m=-1,-2 \Rightarrow c_1e^{-x}+c_2e^{-2x}$. $\text{PI} = \tfrac{1}{2}[1+(3D+D^2)/2]^{-1}(x^3+x^2) = \tfrac{1}{2}[1-z+z^2-z^3](x^3+x^2)$, with $z=(3D+D^2)/2$

$\Rightarrow \text{PI} = \boldsymbol{\frac{1}{2}x^3 - \frac{7}{4}x^2 + \frac{15}{4}x - \frac{31}{8}}$ (checked by substitution)

Example 2 — $(2D^2+2D+3)\,y = x^2+2x-1$

$\text{PI} = \tfrac{1}{3}[1+(2D+2D^2)/3]^{-1}(x^2+2x-1) = \tfrac{1}{3}[x^2+2x-1-(4x+8)/3+8/9] = \boldsymbol{\frac{1}{3}\left(x^2+\frac{2x}{3}-\frac{25}{9}\right)}$. General solution: $y = e^{-x/2}\left(c_1\cos\frac{\sqrt{5}x}{2} + c_2\sin\frac{\sqrt{5}x}{2}\right) + \text{PI}$.

7 PI Type 4 — $X = e^{ax}\cdot V(x)$ (exponential shift)

$$\frac{1}{f(D)}\,e^{ax}V = e^{ax}\,\frac{1}{f(D+a)}\,V$$

Move e^{ax} to the front and replace D by $D + a$. Then use Type 2 (if V = sin/cos) or Type 3 (if $V = x^m$).

BONUS RULE — X = X·V (E.G. X SIN X)

$\frac{1}{f(D)}(xV) = x\,\frac{1}{f(D)}V - \frac{f'(D)}{[f(D)]^2}V$. Alternative: write $x \sin ax = \mathrm{Im}(x\,e^{iax})$ and apply Type 4.

Equation	Working	PI
$(D^2+2)y = x^2e^{3x}$	$e^{3x}\cdot 1/[(D+3)^2+2]\,x^2 = \frac{e^{3x}}{11}[1+(D^2+6D)/11]^{-1}x^2 = \frac{e^{3x}}{11}$ $[x^2-(12x+2)/11+72/121]$	$\frac{e^{3x}}{11}(x^2-\frac{12x}{11}+\frac{50}{121})$
$(D^2-2D+1)y = e^x \sin x$	$e^x\cdot 1/(D+1-1)^2 \sin x = e^x\cdot(1/D^2)\sin x$	$-e^x \sin x$
$(D^2-4D+4)y = x^2e^{2x}$	$e^{2x}\cdot(1/D^2)x^2$ (integrate twice)	$x^4e^{2x}/12$
$(D^2+1)y = x \sin x$	bonus rule / Im of $e^{ix}\cdot 1/(D^2+2iD)\cdot x$	$-(x^2/4)\cos x + (x/4)\sin x$

General solutions from the slides: $(D^2+2)y = x^2e^{3x} \Rightarrow y = c_1\cos\sqrt{2}x + c_2\sin\sqrt{2}x + \mathrm{PI}$; $(D-1)^2y = e^x\sin x \Rightarrow y = (c_1+c_2x)e^x - e^x\sin x$.

8 PI When X Is Any Other Function

Factorise $f(D) = (D-m_1)(D-m_2)\ldots(D-m_n)$ and split into **partial fractions**:

$$\mathrm{PI} = A_1e^{m_1x}\int X\,e^{-m_1x}dx + A_2e^{m_2x}\int X\,e^{-m_2x}dx + \ldots$$

Use this for tan x, sec x, log x, e^{e^x}, $\sin(e^x)$ and similar right-hand sides. For complex roots $\pm ia$, the two terms are conjugates. Substitute $t = e^x$ when e^x appears inside a function.

Equation	Key step	PI
$(D^2+a^2)y = \tan ax$	$\frac{1}{2ia}[\frac{1}{D-ia} - \frac{1}{D+ia}]\tan ax$	$-\frac{1}{a^2}\cos ax\cdot\log(\sec ax + \tan ax)$
$\frac{d^2y}{dx^2} = \ln x$	$1/D^2$: integrate twice, $\int \ln x\,dx = x\ln x - x$	$\frac{x^2\ln x}{2} - \frac{3x^2}{4}$
$(D^2+3D+2)y = e^{e^x}$	$[1/(D+1) - 1/(D+2)]$, with $t = e^x$	$e^{-2x}\,e^{e^x}$
$(D+1)(D+2)y = \sin(e^x)$	$1/(D+1)$ gives $-e^{-x}\cos e^x$; $1/(D+2)$ gives $-e^{-x}\cos e^x + e^{-2x}\sin e^x$	$-e^{-2x}\sin(e^x)$

▸ Which PI rule? — quick decision table

RHS X	Rule	Failure / note
e^{ax}, constant, sinh, cosh, k^x	$D \to a$	$f(a) = 0 \Rightarrow$ multiply by x, differentiate the denominator
$\sin(ax+b)$, $\cos(ax+b)$	$D^2 \to -a^2$	$f(-a^2) = 0 \Rightarrow x \cdot 1/f'(D)$; odd D left $\Rightarrow$ conjugate
x^m, polynomial	$[1 \pm \varphi(D)]^{-1}$ binomial expansion	stop at D^m
$e^{ax}V$	e^{ax} out, $D \to D + a$	then Type 2 or 3 on V
$x\cdot\sin$, $x\cdot\cos$	bonus rule, or Im/Re of e^{iax}	—
tan, sec, cosec, log, e^{e^x}, …	partial fractions + $1/(D-m)$ or **variation of parameters**	2nd-order equations: VOP is usually faster

9 Method of Variation of Parameters (VOP)

VOP gives the PI of $y'' + p\,y' + q\,y = X$ (it also works for variable coefficients) **once the CF is known**.

Step 1: CF $= c_1y_1 + c_2y_2$. **Step 2:** Wronskian $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1y_2' - y_2y_1' \ (\neq 0)$. **Step 3:**

$$\text{PI} = u\,y_1 + v\,y_2, \qquad u = -\int \frac{y_2\,X}{W}dx, \qquad v = \int \frac{y_1\,X}{W}dx$$

The coefficient of y'' must be 1. Divide through first if it is not.

Equation	y_1, y_2, W	Result
$y'' + 4y = \tan 2x$	$\cos 2x$, $\sin 2x$; $W = 2$	$u = -\frac{1}{4}\log(\sec 2x + \tan 2x) + \frac{1}{4}\sin 2x$, $v = -\frac{1}{4}\cos 2x \Rightarrow$ **PI $= -\frac{1}{4}\cos 2x \log(\sec 2x + \tan 2x)$**
$y'' + y = \sec x$	$\cos x$, $\sin x$; $W = 1$	$u = \log\cos x$, $v = x \Rightarrow$ **PI $= \cos x \log\cos x + x \sin x$**
$y'' + y = \operatorname{cosec} x$	$\cos x$, $\sin x$; $W = 1$	**PI $= -x\cos x + \sin x \log\sin x$**
$y'' - 2y' + y = e^x/x$	e^x, $x\,e^x$; $W = e^{2x}$	$u = -x$, $v = \log x \Rightarrow$ **PI $= x\,e^x(\log x - 1)$**

TRAP
Keep the signs straight: the minus sign goes with u (the integral containing y_2). Do not add integration constants. They are already in the CF.

10 Equations with Variable Coefficients — Cauchy & Legendre

A suitable substitution reduces these to equations with constant coefficients. Then use CF and PI exactly as before, but in the variable t.

CAUCHY–EULER EQUATION

$x^n y^{(n)} + k_1 x^{n-1} y^{(n-1)} + \ldots + k_n y = X$

Put $\boldsymbol{x = e^t}$ ($t = \log x$), $\theta = d/dt$:

$xD = \theta$

$x^2D^2 = \theta(\theta - 1)$

$x^3D^3 = \theta(\theta - 1)(\theta - 2)$

LEGENDRE'S LINEAR EQUATION

$k_0(ax + b)^n y^{(n)} + k_1(ax + b)^{n-1} y^{(n-1)} + \ldots + k_n y = X$

Put $\boldsymbol{ax + b = e^t}$, $\theta = d/dt$:

$(ax + b)D = a\theta$

$(ax + b)^2D^2 = a^2\theta(\theta - 1)$

$(ax + b)^3D^3 = a^3\theta(\theta - 1)(\theta - 2)$

Cauchy's equation is Legendre's equation with $a = 1$, $b = 0$. Also convert the RHS: $x = e^t$ (Cauchy), or $x = (e^t - b)/a$ (Legendre). **Write the final answer back in x.**

Example 1 (Cauchy) — $x^2y'' - xy' + y = \log x$

$\theta(\theta - 1) - \theta + 1 = (\theta - 1)^2 \Rightarrow$ CF $= (c_1 + c_2 t)e^t$. PI $= \dfrac{1}{(1-\theta)^2} t = (1 + 2\theta + \ldots)t = t + 2$.

$\Rightarrow \boldsymbol{y = (c_1 + c_2 \log x)x + \log x + 2}$

Example 2 (Legendre) — $(2x + 3)^2y'' - 2(2x + 3)y' - 12y = 6x$

$2x + 3 = e^t$: $4\theta(\theta - 1) - 4\theta - 12 = 4(\theta - 3)(\theta + 1)$; RHS $6x = 3(e^t - 3)$.

CF $= c_1e^{3t} + c_2e^{-t}$; PI $= \dfrac{3}{4}\left[\dfrac{e^t}{-4} - \dfrac{3}{-3}\right] = -\dfrac{3e^t}{16} + \dfrac{3}{4}$

$\Rightarrow \boldsymbol{y = c_1(2x + 3)^3 + \dfrac{c_2}{2x + 3} - \dfrac{3(2x + 3)}{16} + \dfrac{3}{4}}$

COMMON SLIP

In Legendre's equation, **do not forget the powers of a**: $(ax + b)^2D^2 = a^2\theta(\theta - 1)$, not $\theta(\theta - 1)$. With $2x + 3$ the factor is 4.

11 Application 1 — Series LCR Circuit

A resistor R (Ω), an inductor L (H), a capacitor C (F) and an EMF $E(t)$ (V) are connected in series. Q is the charge (C) and $I = dQ/dt$ is the current (A).

Element	Resistor: RI	Inductor: $L\,dI/dt$	Capacitor: Q/C

Kirchhoff's voltage law (sum of voltage drops = applied EMF):

$$L\frac{d^2Q}{dt^2} + R\frac{dQ}{dt} + \frac{Q}{C} = E(t) \qquad \text{or, in terms of current,} \qquad L\frac{d^2I}{dt^2} + R\frac{dI}{dt} + \frac{I}{C} = E'(t)$$

AE: $Lm^2 + Rm + 1/C = 0$. $R^2 > 4L/C$: overdamped; $R^2 = 4L/C$: critically damped; $R^2 < 4L/C$: oscillatory (underdamped). Natural frequency $\omega_0 = 1/\sqrt{(LC)}$.

Example — R = 8 Ω, L = 2 H, C = 0.1 F, E = 10 V, Q(0) = I(0) = 0

$2Q'' + 8Q' + 10Q = 10 \Rightarrow Q'' + 4Q' + 5Q = 5$. AE: $m = -2 \pm i$; CF = $e^{-2t}(c_1 \cos t + c_2 \sin t)$; PI = $5/5 = 1$.

$Q(0) = 0 \Rightarrow c_1 = -1$; $Q'(0) = -2c_1 + c_2 = 0 \Rightarrow c_2 = -2$.

$$\boldsymbol{Q(t) = 1 - e^{-2t}(\cos t + 2 \sin t), \quad I(t) = dQ/dt = 5e^{-2t} \sin t}$$

Example — E = 155 sin 377t, R = 100 Ω, L = 0.1 H, C = 10^{-3} F, q(0) = i(0) = 0

$0.1q'' + 100q' + 1000q = 155 \sin 377t \Rightarrow q'' + 1000q' + 10000q = 1550 \sin 377t$.

AE roots: $r = -500 \pm 200\sqrt{6} \approx -10.10, -989.90$ (overdamped). Try $q_p = A \sin 377t + B \cos 377t \Rightarrow A \approx -0.00128$, $B \approx -0.00366$.

Steady-state current $i_p = q_p' \approx 1.380 \sin 377t - 0.484 \cos 377t$. The transient terms $c_1 e^{r_1 t} + c_2 e^{r_2 t}$ die out; fix c_1, c_2 from the initial conditions.

12 Application 2 — Simple Harmonic Motion (Spring–Mass)

A mass m hangs on a spring with stiffness k. At equilibrium $\boldsymbol{mg = ks}$ (s = static stretch). Let $x(t)$ be the displacement from equilibrium, taking **downward as positive**. Hooke's law gives the restoring force $-k(s + x)$, and Newton's law gives $m\ddot{x} = mg - k(s + x) = -kx$.

$$\frac{d^2x}{dt^2} + \omega^2 x = 0, \qquad \omega^2 = \frac{k}{m} = \frac{g}{s}$$

$$x(t) = c_1 \cos \omega t + c_2 \sin \omega t = A \cos(\omega t - \varphi)$$

Amplitude	Period	Frequency	Max velocity	Phase
$A = \sqrt{(c_1^2 + c_2^2)}$	$T = 2\pi/\omega$	$f = \omega/2\pi = 1/T$	$v_{max} = A\omega$	$\tan \varphi = c_2/c_1$

Units: in the FPS system $g = 32$ ft/s^2, mass = weight/32 (slug), and 1 inch = 1/12 ft.

DAMPED OSCILLATION

With resistance $2\lambda mn\,\dot{x}$ and restoring force mn^2x: $\ddot{x} + 2\lambda n\,\dot{x} + n^2x = 0$, $r = -\lambda n \pm in\sqrt{(1 - \lambda^2)}$ $(0 < \lambda < 1)$.

$x(t) = e^{-\lambda nt}(A \cos \omega t + B \sin \omega t)$, $\omega = n\sqrt{(1 - \lambda^2)}$. With $x(0) = 0$, $\dot{x}(0) = v_0$: $\boldsymbol{x(t) = \frac{v_0}{n\sqrt{(1 - \lambda^2)}}\, e^{-\lambda nt} \sin(n\sqrt{(1 - \lambda^2)}\, t)}$

SHM problem	Solution → answers
12 lb weight stretches the spring 6 in. Pulled 4 in below equilibrium and given an upward velocity of 2 ft/s. No damping.	$m = 12/32 = 3/8$ slug; $k = 12/0.5 = 24$ lb/ft ⇒ $\omega^2 = 64$, $\omega = 8$. $x(0) = 1/3$ ft, $\dot{x}(0) = -2$ ⇒ $c_2 = -1/4$. **$x = \frac{1}{3}\cos 8t - \frac{1}{4}\sin 8t$; $A = 5/12$ ft; $T = \pi/4$ s**
8 lb weight, $k = 12$ lb/ft, raised 5 in above equilibrium and released from rest.	$m = 1/4$ slug ⇒ $\ddot{x} + 48x = 0$, $\omega = 4\sqrt{3}$. $x(0) = -5/12$ ft, $\dot{x}(0) = 0$. **$x = -(5/12)\cos 4\sqrt{3}t$; $A = 5/12$ ft; $T = \pi/(2\sqrt{3})$ s; $f = 2\sqrt{3}/\pi$ Hz; $v_{max} = 5\sqrt{3}/3$ ft/s**

SIGN CONVENTION

Downward is positive. "Raised above equilibrium" means $x(0)$ is negative. "Upward velocity" means $\dot{x}(0)$ is negative. "Released / starts from rest" means $\dot{x}(0) = 0$.

★ One-Page Formula Sheet — Read This Before the Exam

STRUCTURE

$f(D)y = X \Rightarrow y = CF + PI$. AE: $D \to m$.
distinct: $\Sigma c\, e^{mx}$ · repeated: $(c_1 + c_2x)e^{mx}$
$\alpha \pm i\beta$: $e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$

INVERSE OPERATOR

$1/D \to \int \cdot\, 1/(D - a)\; X = e^{ax}\int X\, e^{-ax}dx$

PI RULES

e^{ax}: $D \to a$; if 0, $x \cdot e^{ax}/f'(a)$, $x^2 e^{ax}/f''(a)$
sin/cos ax: $D^2 \to -a^2$; if 0, $x \cdot 1/f'(D)$
$1/(D^2 + a^2)$: sin $ax \to -(x/2a)\cos ax$; cos $ax \to (x/2a)$ sin ax
x^m: $[1 \pm \varphi(D)]^{-1}$ expansion up to D^m
$e^{ax}V$: $e^{ax} \cdot 1/f(D + a) \cdot V$
xV: $x \cdot V/f(D) - f'(D)/f(D)^2 \cdot V$

VARIATION OF PARAMETERS

$W = y_1y_2' - y_2y_1'$
PI $= -y_1\int y_2X/W\, dx + y_2\int y_1X/W\, dx$

CAUCHY / LEGENDRE

$x = e^t$: $xD = \theta$, $x^2D^2 = \theta(\theta - 1)$, $x^3D^3 = \theta(\theta - 1)(\theta - 2)$
$ax + b = e^t$: $(ax + b)^kD^k = a^k\theta(\theta - 1)\ldots(\theta - k + 1)$

LCR

$LQ'' + RQ' + Q/C = E(t)$, $I = Q'$
R^2 vs $4L/C$: over / critical / under-damped

SHM

$\ddot{x} + \omega^2 x = 0$, $\omega = \sqrt{(k/m)} = \sqrt{(g/s)}$
$A = \sqrt{(c_1^2 + c_2^2)}$, $T = 2\pi/\omega$, $f = \omega/2\pi$, $v_{max} = A\omega$
Damped: $\ddot{x} + 2\lambda n\dot{x} + n^2x = 0$

▸ Common mistakes that cost marks

- Writing only the PI. The **general solution is CF + PI**, and the CF must have n constants for an n-th order equation.
- Not spotting the failure case: always check whether a (or $\pm ia$) is a root of the AE **before** applying the shortcut.
- Type 3: expanding past D^m (wasted time) or forgetting the constant factor taken out at the start.
- Cauchy/Legendre: leaving the final answer in t. Convert back using $t = \log x$ or $\log(ax + b)$.
- Applications: using inches instead of feet, and getting the sign of the initial displacement or velocity wrong.

Source: PESU UE26MA141A Unit 2 slides (sessions 1–11) and lecture summaries. Many slide formulas were images, so the worked answers above were re-derived and checked by substitution.